

2012
TRIAL HSC EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen.
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11–14

Total Marks – 70

Section I **10 Marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II **60 Marks**

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section.

Student Number: _____

Teacher: _____

Student Name: _____

QUESTION	MARK
1 – 10	/10
11	/15
12	/15
13	/15
14	/15
TOTAL	/70

Section I**Objective-response questions****Total marks – 10****Attempt Questions 1 – 10**

Answer each question on the multiple choice answer sheet provided.

- 1 The point P divides the interval joining $A(4, 1)$ and $B(-1, 11)$ externally in the ratio $2 : 3$. Which of these are coordinates of P ?

(A) $(2, 5)$ (B) $(14, -19)$ (C) $(1, 7)$ (D) $(-11, 31)$

- 2 What is the sum of the zeroes of $P(x) = 2x^4 + 8x^2 - x + 4$?

(A) 4 (B) -4 (C) 0 (D) -8

- 3 The inverse function of $g(x)$, where $g(x) = \sqrt{2x-4}$ is

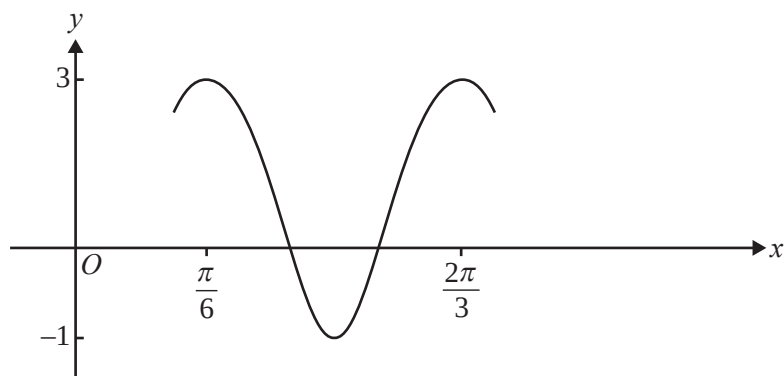
(A) $g^{-1}(x) = \frac{x^2 + 4}{2}$

(B) $g^{-1}(x) = (2x - 4)^2$

(C) $g^{-1}(x) = \sqrt{\frac{x}{2} + 4}$

(D) $g^{-1}(x) = \frac{x^2 - 4}{2}$

- 4 The graph below could have the equation



(A) $y = 2 \cos\left(x + \frac{\pi}{6}\right) + 1$

(B) $y = 2 \cos 2\left(x + \frac{\pi}{6}\right) + 1$

(C) $y = 2 \cos 4\left(x - \frac{\pi}{6}\right) + 1$

(D) $y = 2 \cos 4\left(x + \frac{2\pi}{3}\right) + 1$

- 5 The domain and range of the function $f(x)$, where $f(x) = 3\sin^{-1}(4x-1)$ are respectively.
- (A) $0 \leq x \leq \frac{1}{2}$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ (B) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$.
- (C) $0 \leq x \leq \frac{1}{2}$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$ (D) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$.

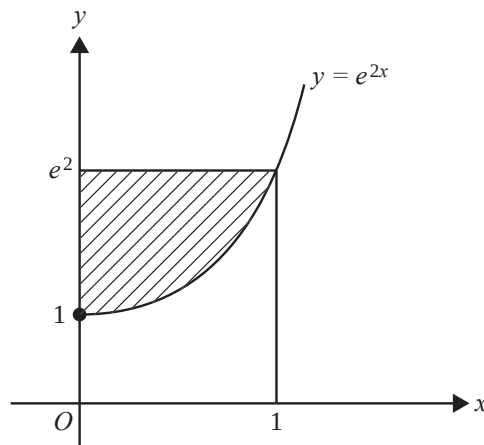
- 6 A particle is moving in simple harmonic motion according to the equation

$$x = 2 - 3\cos\left(2t + \frac{\pi}{3}\right)$$

In which interval does the particle oscillate?

- (A) $-3 \leq x \leq 3$ (B) $\frac{1}{2} \leq x \leq 3\frac{1}{2}$
- (C) $-1 \leq x \leq 5$ (D) $1 \leq x \leq 5$
- 7 $\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right) = ?$
- (A) $-\frac{1}{1+x^2}$ (B) $\frac{1}{1+x^2}$
- (C) $-\frac{x^2}{1+x^2}$ (D) $\frac{x^2}{1+x^2}$

- 8 To find the area of the shaded region in the diagram below, four different students proposed the following calculations.



Student 1: $\int_0^1 e^{2x} dx$

Student 3: $\int_1^{e^2} e^{2y} dy$

Student 2: $e^2 - \int_0^1 e^{2x} dx$

Student 4: $\int_1^{e^2} \frac{\log_e y}{2} dy$

Which of the following is correct?

- (A) Student 2 only. (B) Students 2 and 3 only.
- (C) Students 2 and 4 only. (D) Students 1 and 4 only.

- 9 If the substitution $u = x^2 - 1$ is used then the definite integral $\int_0^2 \frac{x}{\sqrt{x^2 - 1}} dx$

can be simplified to

(A) $\frac{1}{2} \int_{-1}^3 u^{-\frac{1}{2}} du$

(B) $2 \int_{-1}^3 u^{-\frac{1}{2}} du$

(C) $\frac{1}{2} \int_0^2 u^{-\frac{1}{2}} du$

(D) $2 \int_0^2 u^{-\frac{1}{2}} du$

10. The polynomial $P(x)$ is monic and of degree 5. It has a single zero at $x = -2$ and double zero at $x = 1$. Its other two zeroes are not real. Which of the following equations best represents $P(x)$?

(A) $x^5 + bx^4 + cx^3 + dx^2 + ex + f$

(B) $(x-1)(x+2)^2(x+a)(x+b)$

(C) $(x+2)(x-1)^2(x^2+bx+c)$, where $b^2 - 4c \geq 0$.

(D) $(x+2)(x-1)^2(x^2+bx+c)$, where $b^2 - 4c < 0$.

End of Section I

Section II**Free response questions****Total marks – 60****Attempt Questions 11 – 14**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Give that $f(x) = \sin^{-1}(2x)$, find $f''\left(\frac{\sqrt{2}}{4}\right)$ **3**

(b) Find $\int \cos^2 2x \sin 2x \, dx$ **2**

(c) Find the exact value of $\int_0^1 \frac{x^2 + 2}{x^2 + 1} \, dx$ **3**

(d) Evaluate $\lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \sin\left(x - \frac{\pi}{3}\right)}{x - \frac{\pi}{3}}$ **1**

(e) Evaluate $\int_0^1 \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} \, dx$ using the substitution $u = \cos^{-1} x$. **3**

Question 11 continues on the following page

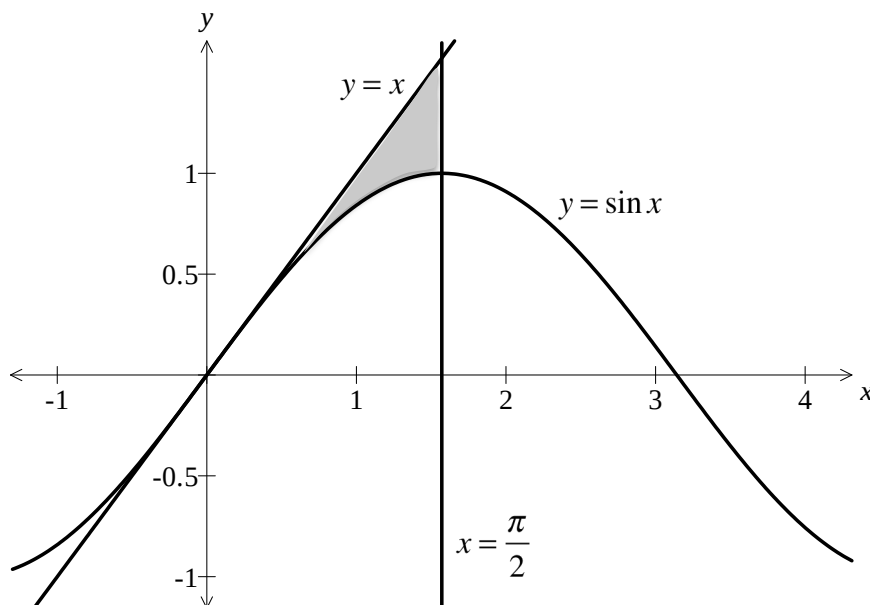
Question 11 continued

- (f) In the diagram below, $y = x$ is the tangent to the curve $y = \sin x$ at $x = 0$.

3

The shaded region is enclosed by $y = x$, $y = \sin x$ and the line $x = \frac{\pi}{2}$.

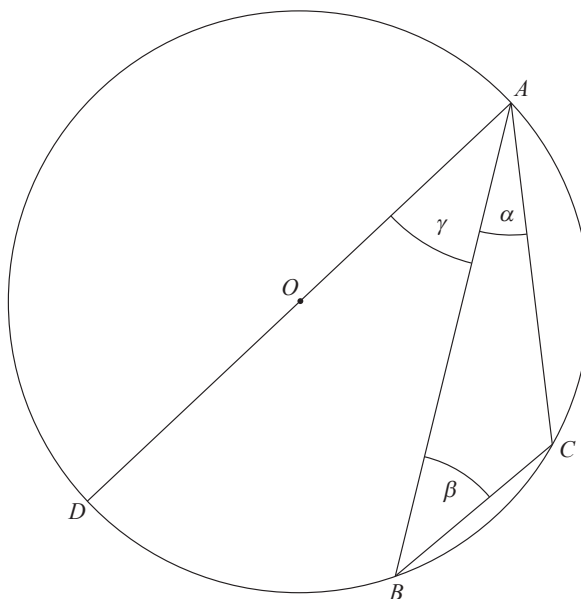
Find the exact volume when the shaded region is rotated about the x -axis.



End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows points B and C on a semi-circle with centre O and diameter AD .



Given that $\angle BAC = \alpha$, $\angle ABC = \beta$ and $\angle OAB = \gamma$,
find the value of $\alpha + \beta + \gamma$, giving reasons.

2

- (b) Two zeroes of the cubic $P(x) = x^3 + px^2 + qx + r$ are equal in magnitude, but opposite in sign.

(i) Show that $x = -p$ is the third zero.

1

(ii) Show that $r = pq$

2

- (c) The Crows Nest Twins Association has 12 members – 6 pairs of twins.

(i) In how many ways can a committee of 3 members be elected if no two members of this committee are each other's twin?

2

(ii) If there are 4 pairs of identical twins and 2 pairs of non-identical twins, in how many ways can a committee of 4 members be elected if there is equal representation of non-identical and identical twins on the committee, but no 2 members of this committee are each other's twin?

2

Question 12 continues on the following page

Question 12 continued

- (d) Falling into cold water is particularly dangerous because the body loses body heat 25 times faster in cold water than in cold air.

The normal body temperature is 37°C .

Shivering begins at an approximate temperature of 36°C , amnesia at 34.4°C and unconsciousness at 30°C .

The body temperature of a person, $T^{\circ}\text{C}$, who has been in the water for t minutes can be modeled by the differential equation

$$\frac{dT}{dt} = k(T - T_w), \text{ where } T_w \text{ is the temperature of the water and } k \text{ is a constant.}$$

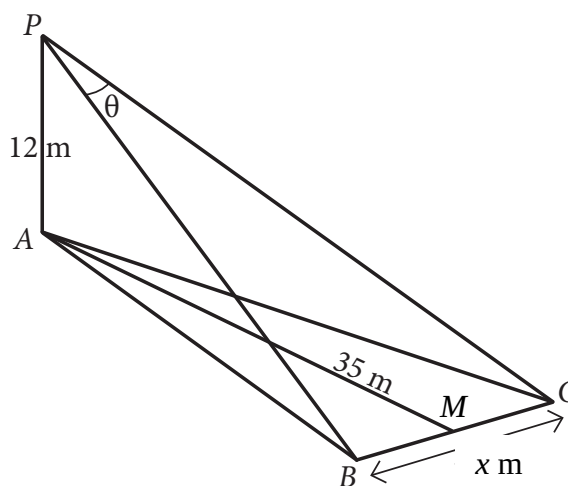
- (i) Show that $T = T_w + Be^{-kt}$, where B is a constant, is a solution of this differential equation. 1
- (ii) If the water temperature is 2°C , a person is expected to start shivering after 5 minutes. 3
Show that $T = 2 + 35e^{-0.0058t}$.
- (iii) Find how long a person can stay in the water before coming unconsciousness. 2
Give your answer in minutes, correct to 1 decimal place.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) By sketching two appropriate graphs, or otherwise, solve $x + 4 > \frac{2}{x+3}$ 3
- (ii) Hence, or otherwise, deduce the values of x for which $x + 4 > \frac{2}{x+3}$ 2

- (b) A hot balloon, P, is rising vertically above A, the vertex of the isosceles triangle ABC. M is the midpoint of BC. $AM = 35$ m and $\angle BPC = \theta$. At a certain time the balloon reaches a height of 12 m above A.

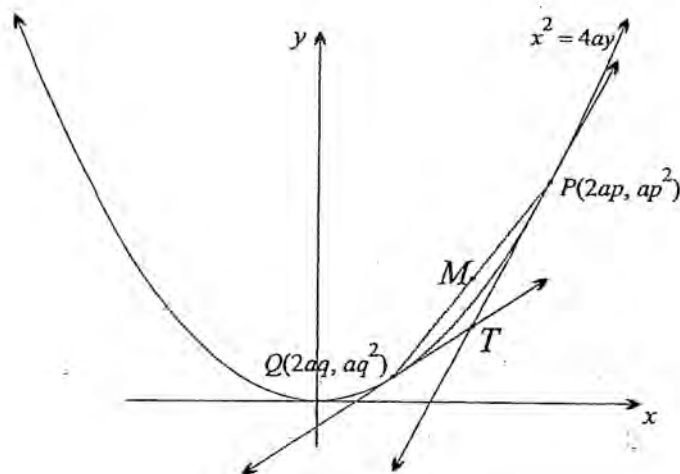


- (i) Find the value of x in terms of θ . 3
- (ii) When $PA = 12$, $\frac{d\theta}{dt} = 0.01$ radians/s, find $\frac{dx}{dt}$ when $\theta = 0.04$ radians. 2
- Leave your answer in correct units, correct to 2 decimal places.

Question 13 continues on the following page

Question 13 continued

- (c) The diagram below shows the tangents drawn at the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$. The tangents at P and Q intersect at T .



You may assume that the equation of the tangent at P is $y = px - ap^2$ and that point T has coordinates $T[a(p+q), apq]$.

- | | | |
|-------|---|----------|
| (i) | Suppose that point T lies on the line $y = a$, show that $pq = 1$. | 1 |
| (ii) | Find the Cartesian equation of the locus of the midpoint, M , of the chord PQ . | 3 |
| (iii) | State any restrictions on the x -coordinate of the locus of M . | 1 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Given that $f(k) = 12^k + 2 \times 5^{k-1}$, 1
show that $f(k+1) - 5f(k) = a \times 12^k$, where a is an integer.
- (ii) Hence, or otherwise, prove by induction that $12^n + 2 \times 5^{n-1}$ is divisible by 7, 3
for all integers $n \geq 1$.
- (b) (i) Use the definition of the derivative i.e. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ 2
to find $f'(0)$ where $f(x) = e^x$.
Hence find $\lim_{h \rightarrow 0} \frac{e^h - 1}{h}$.
- (ii) By considering the sum of a geometric series, find $\lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}}}{n}$ 2
- (c) A particle is moving in simple harmonic motion of period T about a centre O .
Its displacement at any time t is given by $x = A \sin nt$, where A is the amplitude.
- (i) Draw a neat sketch of one period of this displacement-time equation, 2
showing all intercepts.
- (ii) Show that $\dot{x} = \frac{2\pi A}{T} \cos \frac{2\pi t}{T}$ 1
- (iii) The point P lies D units on the positive side of O . 2
Let V be the velocity of the particle when it first passes through P .
Show that the first time the particle is at P after passing through O is
$$\frac{T}{2\pi} \tan^{-1} \left(\frac{2\pi D}{VT} \right)$$
- (iv) Show that the time between the first two occasions when the particle passes 2
through P is $\frac{T}{\pi} \tan^{-1} \frac{VT}{2\pi D}$.
[You may assume that $\tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) = \frac{\pi}{2}$ for $x > 0$]

End of paper

NORTH SYDNEY GIRLS HIGH SCHOOL



2012

TRIAL HSC EXAMINATION

Mathematics Extension 1

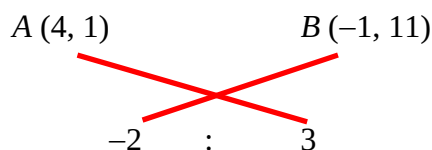
Sample Solutions

Section I**Objective-response Questions**

- | | |
|-----------|---|
| 1 | B |
| 2 | C |
| 3 | A |
| 4 | C |
| 5 | D |
| 6 | D |
| 7 | A |
| 8 | B |
| 9 | A |
| 10 | D |

- 1 The point P divides the interval joining $A(4, 1)$ and $B(-1, 11)$ externally in the ratio $2 : 3$. Which of these are coordinates of P ?

(B) $(14, -19)$



$$x_p = \frac{4 \times 3 + (-1) \times (-2)}{3 + (-2)} = 14; \quad y_p = \frac{1 \times 3 + 11 \times (-2)}{3 + (-2)} = -19$$

$$P(14, -19)$$

- 2 What is the sum of the zeroes of $P(x) = 2x^4 + 8x^2 - x + 4$?

(C) 0

The coefficient of x^3 is zero, so $\sum \alpha = 0$

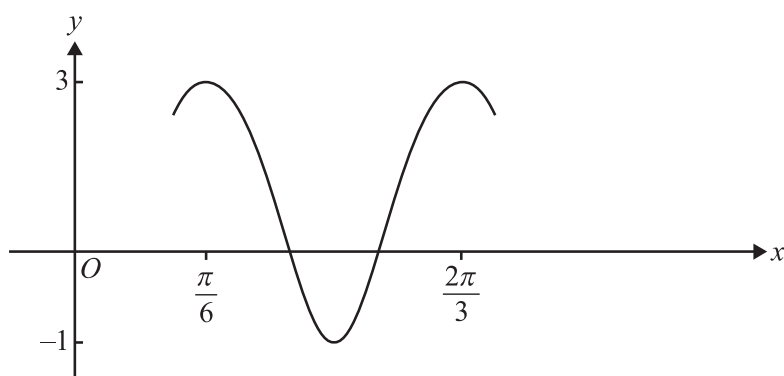
- 3 The inverse function of $g(x)$, where $g(x) = \sqrt{2x-4}$ is

(A) $g^{-1}(x) = \frac{x^2 + 4}{2}$

$$y = \sqrt{2x-4} \Rightarrow x = \sqrt{2y-4}$$

$$\therefore 2y-4 = x^2 \Rightarrow y = \frac{1}{2}(x^2 + 4)$$

- 4 The graph below could have the equation



(C) $y = 1 + 2 \cos 4 \left(x - \frac{\pi}{6} \right)$

Let the period be T

$$T = \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{2}$$

$$T = \frac{2\pi}{n} \Rightarrow n = \frac{2\pi}{T} = 4$$

Also, if the right shift is not obvious then when $x = \frac{\pi}{6}$, $y = 3$.

- 5 The domain and range of the function $f(x)$, where $f(x) = 3\sin^{-1}(4x-1)$ are respectively

(D) $0 \leq x \leq \frac{1}{2}$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

D: $-1 \leq 4x-1 \leq 1$ R: $-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$
 $\therefore 0 \leq 4x \leq 2$ $\therefore -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$
 $\therefore 0 \leq x \leq \frac{1}{2}$

- 6 A particle is moving in simple harmonic motion according to the equation

$$x = 2 - 3\cos\left(2t + \frac{\pi}{3}\right)$$

What is the initial velocity?

(D) $3\sqrt{3}$

$$\dot{x} = 6\sin\left(2t + \frac{\pi}{3}\right)$$

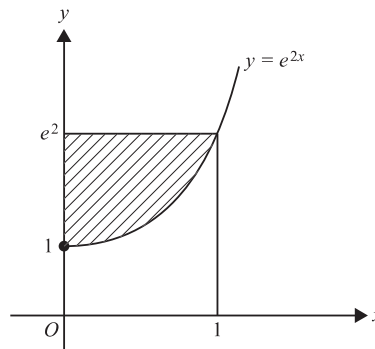
$$t = 0 \Rightarrow \dot{x} = 6\sin\left(\frac{\pi}{3}\right) = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

7 $\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right) = ?$

(A) $-\frac{1}{1+x^2}$

$$\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right) = \frac{1}{1+\frac{1}{x^2}} \times \left(-\frac{1}{x^2}\right) = \frac{x^2}{x^2+1} \times \left(-\frac{1}{x^2}\right) = -\frac{1}{x^2+1}$$

- 8 To find the area of the shaded region in the diagram below, four different students proposed the following calculations.



Student 1: $\int_0^1 e^{2x} dx$

Student 3: $\int_1^{e^2} e^{2y} dy$

Student 2: $e^2 - \int_0^1 e^{2x} dx$

Student 4: $\int_1^{e^2} \frac{\log_e y}{2} dy$

Which of the following is correct?

(B) Students 2 and 4 only.

Student 2 is finding the area next to the x-axis first and then subtracting this from the appropriate rectangle.

Student 4 is finding the area next to the y-axis.

- 9 If the substitution $u = x^2 - 1$ is used then the definite integral $\int_0^2 \frac{x}{\sqrt{x^2 - 1}} dx$ can be simplified to

(A) $\frac{1}{2} \int_{-1}^3 u^{-\frac{1}{2}} du$

$$u = x^2 - 1 \Rightarrow du = 2x$$

$$x = 0 \Rightarrow u = 0 - 1 = -1$$

$$x = 2 \Rightarrow u = 4 - 1 = 3$$

$$\int_0^2 \frac{x dx}{\sqrt{x^2 - 1}} = \frac{1}{2} \int_0^2 \frac{2x dx}{\sqrt{x^2 - 1}} = \frac{1}{2} \int_{-1}^3 u^{-\frac{1}{2}} du$$

- 10 The polynomial $P(x)$ is monic and of degree 5. It has a single zero at $x = -2$ and double zero at $x = 1$. Its other two zeroes are not real. Which of the following expressions best represents $P(x)$?

(D) $(x + 2)(x - 1)^2(x^2 + bx + c)$, where $b^2 - 4c < 0$.

$$\text{It has a single zero at } x = -2 \quad \Rightarrow (x + 2) | P(x)$$

$$\text{Double zero at } x = 1 \quad \Rightarrow (x - 1)^2 | P(x)$$

$$\text{Its other two zeroes are not real} \quad \Rightarrow \Delta < 0$$

Question 11

- (a) Differentiate $\sin^{-1} 2x$ 1
- $$\begin{aligned}\frac{d}{dx}(\sin^{-1} 2x) &= \frac{1}{\sqrt{1-(2x)^2}} \times 2 \\ &= \frac{2}{\sqrt{1-4x^2}}\end{aligned}$$
- (b) Find the acute angle between the lines $y = 3x + 5$ and $2x - 4y + 7 = 0$. 2
 Leave your answer correct to the nearest minute.
 Let the angle be θ
 $m_1 = 3, m_2 = \frac{1}{2}$
- $$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{3 - \frac{1}{2}}{1 + 3 \times \frac{1}{2}} \right| \\ &= 1 \\ \therefore \theta &= 45^\circ\end{aligned}$$
- (c) Find $\int \cos^2 2x \sin 2x \, dx$ 2
- $$\begin{aligned}\int \cos^2 2x \sin 2x \, dx &= -\frac{1}{2} \int \cos^2 2x (-2 \sin 2x) \, dx \\ &= -\frac{1}{2} \int \cos^2 2x \, d \cos 2x \\ &= -\frac{1}{2} \times \frac{1}{3} \cos^3 2x + C \\ &= -\frac{1}{6} \cos^3 2x + C\end{aligned}$$
- (d) Evaluate $\lim_{x \rightarrow 0} \frac{2 \sin x}{3x}$ 1
- $$\lim_{x \rightarrow 0} \frac{2 \sin x}{3x} = \frac{2}{3} \times \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{2}{3} \times 1 = \frac{2}{3}$$
- (e) By splitting the numerator, find the exact value of $\int_0^1 \frac{x^2 + 2}{x^2 + 1} \, dx$ 2
- $$\begin{aligned}\int_0^1 \frac{x^2 + 2}{x^2 + 1} \, dx &= \int_0^1 \frac{(x^2 + 1) + 1}{x^2 + 1} \, dx \\ &= \int_0^1 \left(1 + \frac{1}{x^2 + 1} \right) \, dx \\ &= [x + \tan^{-1} x]_0^1 \\ &= 1 + \frac{\pi}{4}\end{aligned}$$

Question 11 continued

(f) Find $\int_0^1 \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$ using the substitution $u = \cos^{-1} x$. 3

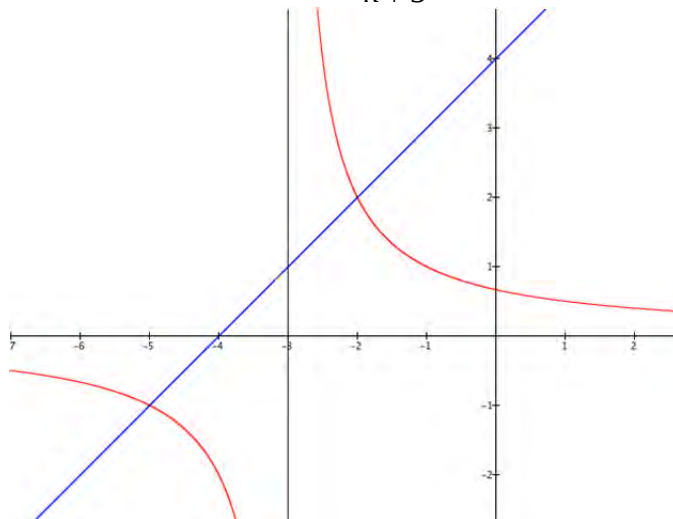
$$u = \cos^{-1} x \Rightarrow du = -\frac{1}{\sqrt{1-x^2}} dx$$

$$x = 0 \Rightarrow u = \cos^{-1} 0 = \frac{\pi}{2}$$

$$x = 1 \Rightarrow u = \cos^{-1} 1 = 0$$

$$\begin{aligned} \int_0^1 \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx &= -\int_0^1 e^{\cos^{-1} x} \times \frac{-1}{\sqrt{1-x^2}} dx \\ &= -\int_{\frac{\pi}{2}}^0 e^u du = \int_0^{\frac{\pi}{2}} e^u du \\ &= [e^u]_0^{\frac{\pi}{2}} \\ &= e^{\frac{\pi}{2}} - 1 \end{aligned}$$

(g) (i) By sketching two appropriate graphs on the same number plane, 3
or otherwise, solve $x + 4 > \frac{2}{x+3}$.



Finding the points of intersection: $x + 4 = \frac{2}{x+3}$

$$\therefore x^2 + 7x + 12 = 2$$

$$\therefore x^2 + 7x + 10 = 0$$

$$\therefore (x+2)(x+5) = 0$$

$$\therefore x = -2, -5$$

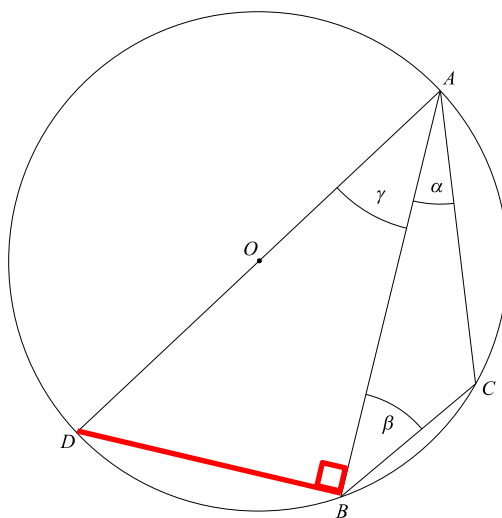
$$\therefore -5 < x < -3, x > -2$$

(ii) Hence, or otherwise, deduce the values of x for which $x + 4 > \frac{2}{x+3}$. 1

From the diagram, $x > -2$.

Question 12

- (a) The diagram shows points B and C on a semi-circle with centre O and diameter AD .



Given that $\angle BAC = \alpha$, $\angle ABC = \beta$ and $\angle OAB = \gamma$,
find the value of $\alpha + \beta + \gamma$, giving reasons.

2

Join D to B

$$\angle ADB = 90^\circ \quad (\text{angle in a semi-circle})$$

$$\angle DAC + \angle DBC = 180^\circ \quad (\text{opp. ang. of cyclic quad})$$

$$\therefore \alpha + \gamma + 90^\circ + \beta = 180^\circ$$

$$\therefore \alpha + \beta + \gamma = 90^\circ$$

- (b) Two zeroes of the cubic polynomial $P(x) = x^3 + px^2 + qx + r$ are equal in magnitude, but opposite in sign.

- (i) Show that $x = -p$ is the third zero.

1

Let the three roots be α , $-\alpha$ and β .

$$\sum \alpha = -p$$

$$\therefore \alpha + (-\alpha) + \beta = -p$$

$$\therefore \beta = -p$$

- (ii) Show that $r = pq$

2

$$-\alpha^2 \beta = -r$$

$$\therefore \alpha^2 \beta = r$$

$$\alpha \beta + (-\alpha) \beta + (-\alpha) \alpha = q$$

$$\therefore -\alpha^2 = q \Rightarrow \alpha^2 = -q$$

$$\therefore r = \alpha^2 \beta = -q \times (-p) = pq$$

ALTERNATIVE:

$$\begin{aligned} P(-p) &= (-p)^3 + p(-p)^2 + q(-p) + r \\ &= -pq + r \end{aligned}$$

$$P(-p) = 0 \quad [\text{From (i)}]$$

$$\therefore r = pq$$

Question 12 continued

(c) The Crows Nest Twins Association has 12 members i.e. 6 pairs of twins.

- (i) *In how many ways can a committee of 3 members be elected if no two members of this committee are each other's twin?* 2

Let the twins be Aa, Bb, Kk, Dd, Ee, Ff

First choose 3 sets of twins in ${}^6C_3 = 20$ ways.

Then from each set, there is a choice of choosing either twin.

So total number of ways is ${}^6C_3 \times 2 \times 2 \times 2 = 160$

ALTERNATIVE:

Pick 1 person from the 12 possible, then to avoid choosing the person's twin there are only 10 possibilities to pick the next person and then 8 for the final choice i.e. $12 \times 10 \times 8 = 960$. But this means that if we picked **A**, **B** and **d** then this will also end up giving us **d**, **A** and **B** and all the other permutations of **A**, **B** and **d**.

So to avoid this over counting, the number of possible committees is $\frac{12 \times 10 \times 8}{3!} = 160$.

NB we wouldn't divide by 3! if we were picking President, VP and Secretary.

- (ii) *If there are 4 pairs of identical twins and 2 pairs of non-identical twins, in how many ways can a committee of 4 members be elected if there is equal representation of non-identical and identical twins on the committee, but no 2 members of this committee are each other's twin?* 2

Let the identical twins be Aa, Bb, Kk, Dd and the non-identical twins be Ee, Ff. There has to be 2 identical twins and 2 non-identical twins.

First choose 2 sets of identical twins in ${}^4C_2 = 6$ ways.

Then from each set, there is a choice of choosing either twin.

So total number of ways of choosing the identical twins is ${}^4C_2 \times 2 \times 2 = 24$

Now choosing 2 sets of non-identical twins can be done in ${}^2C_2 = 1$ ways.

Then from each set, there is a choice of choosing either twin

So total number of ways of choosing the non-identical twins is ${}^2C_2 \times 2 \times 2 = 4$.

So choosing the committee can be done in $({}^4C_2 \times 2^2) \times ({}^2C_2 \times 2^2) = 96$ ways.

ALTERNATIVE:

Using a similar logic to above and starting with the identical twins:

Pick 1 person from the 8 possible, then to avoid choosing the person's twin there are only 6 possibilities to pick the next person i.e. $8 \times 6 \times 4 = 48$.

But this means that if we picked **A** and **B** then this will also end up giving us **B** and **A**.

So to avoid this over counting the number of possible committees is $\frac{8 \times 6}{2!} = 24$.

Now with the non-identical twins:

Pick 1 person from the 4 possible, then to avoid choosing the person's twin there are only 2 possibilities to pick the next person i.e. $4 \times 2 = 8$.

But this means that if we picked **E** and **F** then this will also end up giving us **F** and **E** $\frac{4 \times 2}{2!} = 4$.

As we want both of these groups on the committee then there are $24 \times 4 = 96$ ways of this.

- (d) Falling into cold water is particularly dangerous because the body loses body heat 25 times faster in cold water than in cold air.

The normal body temperature is 37°C .

Shivering begins at an approximate temperature of 36°C , amnesia at 34.4°C and unconsciousness at 30°C .

The body temperature of a person, $T^{\circ}\text{C}$, who has been in the water for t minutes can be modeled by the differential equation

$$\frac{dT}{dt} = -k(T - W), \text{ where } W \text{ is the temperature of the water and } k \text{ is a constant.}$$

- (i) Show that $T = W + Be^{-kt}$, where B is a constant, is a solution of this differential equation. 1

$$T = W + Be^{-kt} \Rightarrow T - W = Be^{-kt}$$

$$\text{LHS} = \frac{dT}{dt} = -kBe^{-kt} = -k(T - W) = \text{RHS}$$

- (ii) If the water temperature is 2°C , a person is expected to start shivering after 5 minutes. 3

$$\text{Show that } T = 2 + 35e^{-0.0058t}.$$

$$W = 2 \Rightarrow T = 2 + Be^{-kt}$$

$$t = 5, T = 37^{\circ} \Rightarrow T = 2 + 35e^{-kt}$$

$$t = 5, T = 36^{\circ}:$$

$$\therefore 36 = 2 + 35e^{-5k} \Rightarrow 35e^{-5k} = 34$$

$$\therefore e^{-5k} = \frac{34}{35} \Rightarrow -5k = \ln \frac{34}{35}$$

$$\therefore k = \frac{1}{5} \ln \frac{35}{34} \doteq 0.0058$$

- (iii) Find how long a person can stay in the water before becoming unconsciousness. 2
Give your answer in minutes, correct to 1 decimal place.

Unconsciousness at 30°C .

$$T = 2 + 35e^{-0.0058t}$$

$$\therefore 30 = 2 + 35e^{-0.0058t}$$

$$\therefore 35e^{-0.0058t} = 28$$

$$\therefore e^{-0.0058t} = \frac{28}{35} = \frac{4}{5}$$

$$\therefore -0.0058t = \ln \frac{4}{5}$$

$$\therefore t = \frac{\ln \frac{5}{4}}{0.0058} \doteq 38.5$$

So unconsciousness would occur after approximately 38.5 mins.

Question 13

- (a) When the region is rotated about the x -axis, the volume of the solid formed is given by the integral $\pi \int_0^{\frac{\pi}{2}} (x^2 - \sin^2 x) dx$.

Find the exact volume, leaving your answer in exact, simplified form.

3

$$\cos 2x = 1 - 2\sin^2 x \Rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\begin{aligned} \pi \int_0^{\frac{\pi}{2}} (x^2 - \sin^2 x) dx &= \pi \int_0^{\frac{\pi}{2}} \left[x^2 - \frac{1}{2}(1 - \cos 2x) \right] dx \\ &= \pi \left[\frac{x^3}{3} - \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \right]_0^{\frac{\pi}{2}} \\ &= \pi \left[\frac{\left(\frac{\pi}{2}\right)^3}{3} - \frac{1}{2} \left(\frac{\pi}{2} - \frac{1}{2} \sin \pi \right) \right] \\ &= \pi \left(\frac{\pi^3}{24} - \frac{\pi}{4} \right) \\ &= \frac{\pi^2(\pi^2 - 6)}{24} \end{aligned}$$

- (b) Show that there is a stationary point on the curve $y = e^{-x} - e^{-2x}$ at $x = \ln 2$ and determine its nature.

3

$$e^{\ln x} = x \text{ and } e^{-\ln x} = \frac{1}{x}$$

$$y = e^{-x} - e^{-2x}$$

$$\therefore y' = -e^{-x} + 2e^{-2x}$$

$$\therefore y'' = e^{-x} - 4e^{-2x}$$

$$x = \ln 2 \Rightarrow y' = -e^{-\ln 2} + 2e^{-2\ln 2} = -\frac{1}{2} + 2 \times \frac{1}{4} = 0$$

So there is a stationary point at $x = \ln 2$

$$x = \ln 2 \Rightarrow y'' = e^{-\ln 2} - 4e^{-2\ln 2} = \frac{1}{2} - 4 \times \frac{1}{4} = -\frac{1}{2}$$

As $y'' < 0$, the stationary point is a (rel) maximum turning point.

OR

$$x = \ln 2 \div 0.693$$

x	0.6	$\ln 2$	0.7
y'	0.054	0	-0.0034
	/	-	\

So there is a (rel) maximum turning point at $x = \ln 2$.

Question 13 continued

- (c) The diagram below shows the tangents drawn at the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$.

The tangents at P and Q intersect at T .

You may assume that the equation of the tangent at P is $y = px - ap^2$ and that point T has coordinates $T[a(p+q), apq]$.

- (i) Supposing that point T lies on the line $y = a$, show that $pq = 1$. 1
If point T lies on the line $y = a$, then $apq = a \Rightarrow pq = 1$

- (ii) M is the midpoint of the chord PQ .
Find the Cartesian equation of the locus of M . 3

$$M\left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2}\right) = \left[a(p+q), \frac{a}{2}(p^2+q^2)\right]$$

$$\text{Let } x = a(p+q) \text{ and } y = \frac{a}{2}(p^2+q^2)$$

$$x = a(p+q) \Rightarrow p+q = \frac{x}{a}$$

$$y = \frac{a}{2}(p^2+q^2) \Rightarrow p^2+q^2 = \frac{2y}{a}$$

$$\frac{2y}{a} = p^2+q^2$$

$$= (p+q)^2 - 2pq$$

$$= \left(\frac{x}{a}\right)^2 - 2$$

$$\therefore \frac{2y}{a} = \left(\frac{x}{a}\right)^2 - 2$$

$$\therefore x^2 = 2a(y+a) \text{ or } y = \frac{1}{2a}x^2 - a$$

- (iii) State any restrictions on the domain of the locus of M . 1

$$x^2 = 4ay \Rightarrow y = \frac{1}{4a}x^2$$

M is constrained in its domain since T has to lie on the line $y = a$ and *external* to the curve as tangents to a parabola can only intersect externally.

So the constraints for T will be the restraints for M .

Comparing y -coordinates, M must be above the parabola and T must be below it

$$\text{i.e. } y = \frac{1}{4a}x^2 > a$$

$$\therefore x^2 > 4a^2$$

$$\therefore x < -2a, x > 2a \text{ or } |x| > 2a$$

Alternative 1:

Comparing y -coordinates, M must be above the parabola.

$$\text{i.e. } y = \frac{1}{2a}x^2 - a > a$$

$$\therefore \frac{1}{2a}x^2 > 2a$$

$$\therefore x^2 > 4a \Rightarrow x < -2a, x > 2a$$

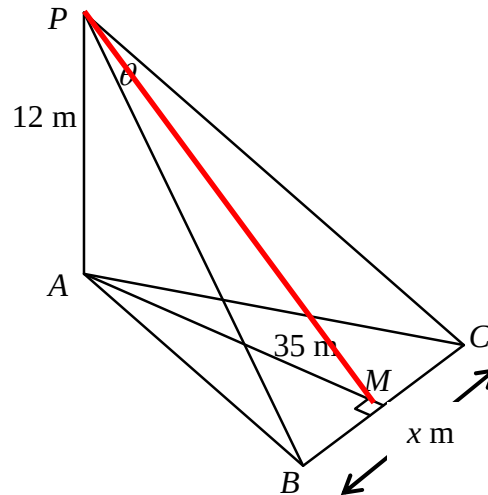
Alternative 2:

As M and T share the same x -coordinate, the extreme case will be when P and Q are NOT distinct points i.e. $p = q$.

When this happens T (and M) will be at $(2a, a)$.

Since T must lie outside the parabola then $|x| > 2a$.

- (d) A hot air balloon P is rising vertically above A and P is equidistant from observers at B and C . M is the midpoint of BC , $AM = 35$ m and $\angle BPC = \theta$. At a certain time the balloon reaches a height of 12 m above A .



- (i) Show that $x = 74 \tan\left(\frac{\theta}{2}\right)$

2

$$PM = 37 \quad (\text{Pythagorean triad})$$

As $\triangle PBC$ is isosceles then $\angle MPB = \frac{1}{2}\theta$ and $BM = \frac{1}{2}x$

$$\therefore \tan\left(\frac{1}{2}\theta\right) = \frac{\frac{1}{2}x}{37}$$

$$\therefore x = 74 \tan\left(\frac{1}{2}\theta\right)$$

- (ii) When $PA = 12$, $\frac{d\theta}{dt} = 0.01$ radians/s. Find $\frac{dx}{dt}$ when $\theta = 0.04$ radians.

2

Leave your answer correct to 2 decimal places.

$$\frac{dx}{d\theta} = \frac{1}{2} \times 74 \times \sec^2\left(\frac{1}{2}\theta\right) = 37 \sec^2\left(\frac{1}{2}\theta\right)$$

$$\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$= 37 \sec^2(0.02) \times 0.01$$

$$= 0.37 \text{ m/s}$$

Question 14

- (a) Prove by induction that $12^n + 2 \times 5^{n-1}$ is divisible by 7, for all integers $n \geq 1$.

3

$$\text{Test } n = 1: \quad 12^n + 2 \times 5^{n-1} = 12 + 2 \times 5 = 12 + 10 = 22 = 2 \times 11.$$

So true for $n = 1$.

$$\text{Assume true for } n = k \text{ i.e. } 12^k + 2 \times 5^{k-1} = 7M, M \in \mathbb{Z}$$

$$\therefore 12^k = 7M - 2 \times 5^{k-1}$$

$$\text{Need to prove true for } n = k + 1 \text{ i.e. } 12^{k+1} + 2 \times 5^k = 7N, N \in \mathbb{Z}$$

$$\begin{aligned} 12^{k+1} + 2 \times 5^k &= 12 \times 12^k + 2 \times 5^k \\ &= 12(7M - 2 \times 5^{k-1}) + 2 \times 5 \times 5^{k-1} \\ &= 7 \times 12M - 24 \times 5^{k-1} + 10 \times 5^{k-1} \\ &= 7(12M - 2 \times 5^{k-1}) \\ &= 7N \quad \left(N = 12M - 2 \times 5^{k-1} \in \mathbb{Z} \right) \end{aligned}$$

So the statement is true for $n = k + 1$ if it is true for $n = k$.

So by the principle of mathematical induction the statement is true for all integers $n \geq 1$.

Question 14 continued

$$(b) \quad (i) \quad f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^h - 1}{h}.$$

$$\text{Now } f'(x) = e^x$$

$$\therefore f'(0) = 1$$

$$\therefore f'(0) = \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

$$(ii) \quad e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}} = \frac{e^{\frac{1}{n}} \left[\left(e^{\frac{1}{n}} \right)^n - 1 \right]}{\frac{1}{n} - 1}$$

$$= \frac{e^{\frac{1}{n}} (e - 1)}{e^{\frac{1}{n}} - 1}$$

$$(iii) \quad \text{Using (i), if } h = e^{\frac{1}{n}}, \text{ then as } n \rightarrow \infty, h = \frac{1}{n} \rightarrow 0.$$

$$\text{NB } \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = \lim_{h \rightarrow 0} \frac{h}{e^h - 1} = 1$$

From (ii) and replacing n with $\frac{1}{h}$.

$$\frac{e^{\frac{1}{n}}}{n \left(e^{\frac{1}{n}} - 1 \right)} (e - 1) = \frac{e^h}{\frac{1}{h} (e^h - 1)} (e - 1) = \frac{he^h}{(e^h - 1)} (e - 1)$$

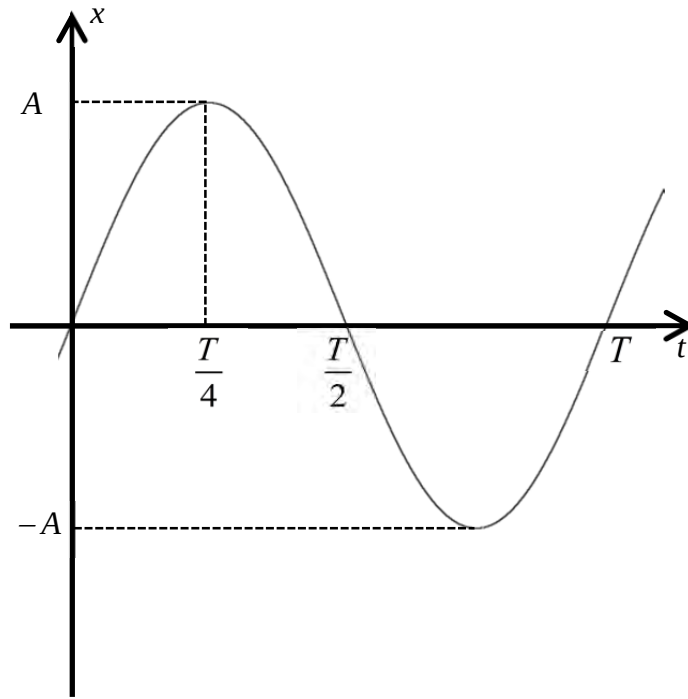
$$\therefore \lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}}}{n} = (e - 1) \times \lim_{h \rightarrow 0} \left[\frac{h}{(e^h - 1)} \right] \times \lim_{h \rightarrow 0} e^h$$

$$= (e - 1) \times 1 \times 1$$

$$= e - 1$$

Question 14 continued

(c) (i)



$$(ii) \quad T = \frac{2\pi}{n} \Rightarrow n = \frac{2\pi}{T}$$

$$x = A \sin nt = A \sin \left(\frac{2\pi}{T} \right) t$$

$$\therefore \dot{x} = A \times \frac{2\pi}{T} \cos \left(\frac{2\pi}{T} \right) t = \frac{2\pi A}{T} \cos \left(\frac{2\pi}{T} \right) t$$

$$(iii) \quad A, D > 0$$

At P, $x = D$, and $\dot{x} = V$

$$D = A \sin \left(\frac{2\pi}{T} \right) t \quad -(1)$$

$$V = \frac{2\pi A}{T} \cos \left(\frac{2\pi}{T} \right) t \quad -(2)$$

$$(1) \div (2) \Rightarrow \frac{T}{2\pi} \tan \left(\frac{2\pi}{T} \right) t = \frac{D}{V}$$

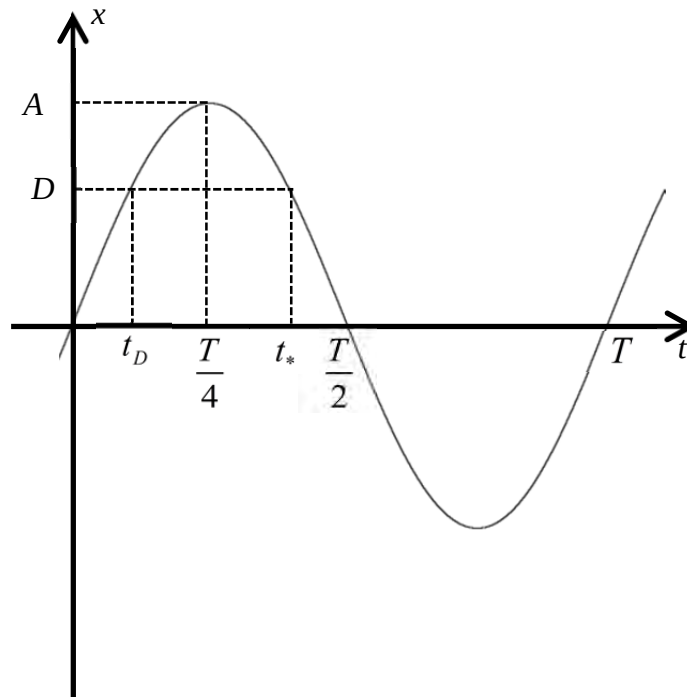
$$\therefore \tan \left(\frac{2\pi}{T} \right) t = \frac{2\pi D}{VT}$$

As D , V and T are positive, then the first positive solution is $t = \frac{T}{2\pi} \tan^{-1} \left(\frac{2\pi D}{VT} \right)$

Question 14 continued

(iv) **Method 1:** Using (i)

Let $t_D = \frac{T}{2\pi} \tan^{-1}\left(\frac{2\pi D}{VT}\right)$ and let t_* be the second time that the particle is at P



Using the symmetry of the sine curve about $t = \frac{T}{4}$, for $0 \leq t \leq \frac{T}{2}$ then

$$\begin{aligned}
 t_* - t_D &= \frac{T}{2} - 2t_D \\
 &= \frac{T}{2} - \frac{T}{\pi} \tan^{-1}\left(\frac{2\pi D}{VT}\right) \\
 &= \frac{T}{\pi} \left[\frac{\pi}{2} - \tan^{-1}\left(\frac{2\pi D}{VT}\right) \right] \\
 &= \frac{T}{\pi} \tan^{-1}\left(\frac{VT}{2\pi D}\right) \qquad \left[\tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \tan^{-1} x \right]
 \end{aligned}$$

Question 14 continued

Method 2: Using SHM

Given that the particle is travelling in SHM, then the velocity of the particle when it comes back to P is $-V$

$$D = A \sin\left(\frac{2\pi}{T}t\right) \quad -(1)$$

$$-V = \frac{2\pi A}{T} \cos\left(\frac{2\pi}{T}t\right) \quad -(2)$$

$$(1) \div (2) \Rightarrow \frac{T}{2\pi} \tan\left(\frac{2\pi}{T}t\right) = -\frac{D}{V}$$

$$\therefore \tan\left(\frac{2\pi}{T}t\right) = -\frac{2\pi D}{VT}$$

$$\text{As } D, V \text{ and } T \text{ are positive then } \frac{2\pi t}{T} = \pi - \tan^{-1}\left(\frac{2\pi D}{VT}\right)$$

$$\therefore t = \frac{T}{2} - \frac{T}{2\pi} \tan^{-1}\left(\frac{2\pi D}{VT}\right)$$

$$\therefore t = \frac{T}{\pi} \left[\frac{\pi}{2} - \tan^{-1}\left(\frac{2\pi D}{VT}\right) \right]$$

$$\therefore t = \frac{T}{\pi} \tan^{-1}\left(\frac{VT}{2\pi D}\right) \quad \left[\tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \tan^{-1}x \right]$$